

AI-Native Multicast Routing in FANETs using Graph Neural Networks and Deep Hierarchical Reinforcement Learning for Disaster Recovery Video Streaming

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Abstract—Flying Ad Hoc Networks (FANETs) present unique routing challenges due to rapid three-dimensional mobility, frequent topology changes, and strict energy and latency constraints. Traditional MANET routing and bio-inspired optimizers such as Artificial Bee Colony (ABC/AABC) address some routing objectives but struggle to adapt online to the extreme dynamics of UAV swarms. This paper proposes a Multi-Agent Deep Reinforcement Learning (MADRL) based framework for energy-aware multicast routing in FANETs. Each UAV is modeled as an autonomous learning agent that observes local link metrics, residual energy, and neighbor states, and selects forwarding actions to build and maintain multicast distribution trees. A centralized training with a decentralized execution paradigm (centralized critic / decentralized actors) is used to stabilize learning while preserving distributed deployability. The framework integrates a graph-aware state representation, an adaptive reward that balances packet delivery ratio, latency, and energy consumption, and an online route maintenance module that replans under link failures. We implement the approach in a simulation pipeline (ns-3 + a custom Gym environment + PyTorch MADRL models) and compare it against AABC-based routing and classical protocols (AODV, OLSR). Simulation results demonstrate improved packet delivery ratio, reduced end-to-end delay, and extended network lifetime under high mobility scenarios. The proposed MADRL multicast solution shows promise for robust FANET routing where adaptivity and energy efficiency are critical.

Index Terms—Flying Ad Hoc Networks, FANET, Multi-Agent Deep Reinforcement Learning, Multicast Routing, Energy-Aware Routing, Centralized Critic.

I. INTRODUCTION

Flying Ad Hoc Networks (FANETs) formed by cooperating unmanned aerial vehicles (UAVs) are increasingly used in applications such as disaster response, environmental monitoring, precision agriculture, and tactical communications. Compared to ground MANETs, FANETs operate in three dimensions, have higher node speeds, and face more frequent topology changes, which significantly reduces route lifetime and complicates multicast distribution to multiple receivers. Standard MANET routing protocols (e.g., AODV, OLSR) and swarm-inspired optimizers (e.g., Artificial Bee Colony variants) can provide baseline functionality, but they are often brittle under extreme mobility and do not learn to trade off multiple competing QoS objectives online. Recent surveys highlight these FANET challenges and call for adaptive, learning-based routing solutions that exploit local observations to make robust forwarding decisions.

This work proposes an energy-aware MADRL framework for multicast routing in FANETs that learns distributed forwarding policies. Instead of static route discovery and periodic flooding, agents learn to choose next hops that maximize long-term multicast delivery while minimizing delay and energy drain. We adopt centralized training with decentralized execution (CTDE) to provide a stable learning signal (centralized critic) while enabling each UAV to operate on local observations at runtime. Our contributions are: (1) a problem formulation of multicast routing in 3D FANETs as a cooperative multi-agent Markov decision process (Dec-POMDP) with a custom reward balancing delivery, latency, and energy; (2) a graph-aware state encoding that captures local topology and link quality; (3) an MADRL training pipeline and route maintenance module suitable for high mobility; and (4) simulation results showing gains over AABC-based routing and classic MANET protocols.

II. LITERATURE SURVEY

Routing in FANETs has been studied extensively; surveys emphasize route instability due to high mobility, limited energy of onboard UAVs, and the need for three-dimensional mobility models and link quality estimation. Approaches range from topology-based protocols adapted from MANETs to hierarchical and geographic routing tailored to FANET use cases.

Swarm intelligence and bio-inspired metaheuristics (e.g., Artificial Bee Colony and its adaptive variants) have been applied to MANET routing to optimize path selection with objectives such as hop count and residual energy. These methods produce good offline or slowly changing solutions but require frequent re-optimization in rapidly changing networks, which undermines responsiveness in FANETs.

Reinforcement learning (RL), and particularly multi-agent deep RL (MADRL), has recently been explored for networking tasks where dynamic adaptation is advantageous. Prior works demonstrate MADRL for offloading decisions, traffic scheduling, and delay-constrained resource allocation; several studies specifically propose MAPPO/MADDPG variants for intra-FANET decision problems (e.g., scheduling and offloading). These results suggest MADRL can handle the combinatorial and non-stationary aspects of FANET routing when combined with suitable state encodings and training regimes.

Taken together, prior art motivates a multicast routing design that combines the adaptivity of MADRL, the distributed requirements of UAV swarms, and the energy/latency tradeoffs essential for practical FANET deployments.

III. SYSTEM ARCHITECTURE

We design a layered, modular architecture that separates simulation/control, agent learning, and runtime packet-forwarding components to enable both research and prototyping (Fig. 1 — conceptual).

A. Network and Simulation Layer

Defines the FANET physical and MAC layer behavior used during training and evaluation. UAV mobility follows configurable 3D models (e.g., 3D Random Waypoint, Gauss-Markov); radio propagation includes path loss, shadowing, and simple fading. The layer is implemented in ns-3 (for

accurate link behavior) and connected to a Gym-compatible environment for RL training.

B. Agent Layer (Runtime/Execution)

Each UAV hosts an actor policy that takes local observations and outputs forwarding actions for multicast packets (which neighbor(s) to forward to, or whether to aggregate/drop). Actors run lightweight inference at runtime.

C. Centralized Critic / Training Layer

During training, a centralized critic (accessible only in simulation) observes joint states (or summaries) to compute a stable value estimate and guide actor updates (CTDE). Training uses PyTorch-based MADRL algorithms (e.g., MAPPO / MADDPG variants) and experience replay tailored for non-stationary multi-agent settings.

D. Routing & Maintenance Layer

Contains multicast group management, tree construction heuristics (seeded by agents' actions), and failure recovery modules that trigger rapid local replanning when link breakages are detected.

E. Monitoring & Replay Layer

Collects trace logs (packet deliveries, latencies, energy consumption) for offline analysis and for building curriculum schedules that expose agents to increasing mobility and density conditions.

IV. PROPOSED METHODOLOGY

We cast multicast routing as a cooperative Dec-POMDP. Below we detail state, action, reward, training algorithm, and route maintenance.

A. Problem Formulation (Dec-POMDP)

The multicast routing problem in Flying Ad Hoc Networks (FANETs) is modeled as a **Decentralized Partially Observable Markov Decision Process (Dec-POMDP)**. In this formulation, each UAV operates as an autonomous agent that makes routing decisions based on its local observations while cooperating with other agents to achieve efficient multicast communication.

Agents

The system consists of **N UAVs**, each represented as an agent indexed by **i**, where $i \in \{1, 2, \dots, N\}$. Each UAV is capable of sensing its local environment, communicating with neighboring UAVs, and forwarding packets to maintain multicast connectivity.

Joint State

The global state at time **t**, denoted as **s(t)**, represents the overall network condition and includes:

Positions of all UAVs in three-dimensional space

Residual energy levels of each UAV

Link quality measurements between neighboring nodes

Multicast group membership information

During the training phase, this joint state information is accessible to the **centralized critic** to improve learning stability. However, during execution each UAV relies only on its local observations.

Local Observation

Each agent **i** observes a partial view of the environment, represented as **o_i(t)**. The local observation includes:

Current 3D position of UAV **i**, represented as **p_i(t)**

Residual energy level of UAV **i**, represented as **E_i(t)**

Neighbor list with estimated link quality values **L_{ij}** for each neighbor **j**

Information about buffered multicast packets such as **Time-To-Live (TTL)** and **sequence identifiers**

These observations allow each UAV to make routing decisions based on its immediate network surroundings.

Action Space

At each time step, an agent selects an action **a_i(t)** from a discrete action set **A**. For each incoming multicast packet, the UAV chooses one of the following forwarding decisions:

A = {
Forward packet to neighbor **j**,
Aggregate packet and forward,
Buffer packet temporarily,
Drop packet
}

In extended system variants, UAVs may also perform **mobility-assisting actions**, such as small course adjustments to improve connectivity and maintain stable communication links.

B. State Encoding (Graph-Aware Representation)

Since FANETs have a dynamic topology with varying numbers of neighboring nodes, the local observations are encoded using a **graph-aware representation**. This encoding allows the model to handle variable network structures effectively.

For each neighboring UAV **j**, a feature vector is constructed:

Neighbor Vector = [delta_p_ij, L_ij, E_j]

Where:

delta_p_ij represents the relative position between UAV **i** and UAV **j**

L_ij represents the link quality between UAV **i** and UAV **j**

E_j represents the residual energy of neighbor **j**

These neighbor vectors are aggregated using permutation-invariant operations such as **summation or attention mechanisms** to produce a fixed-length embedding. This embedding captures the local topology and enables the learning model to generalize across different network sizes and densities.

C. Reward Design

The reward function is designed to encourage efficient routing behavior while minimizing unnecessary energy consumption and network congestion.

The reward for agent **i** at time **t** is defined as:

$r_i(t) = \alpha * \text{delta_PDR}(t) - \beta * \text{delta_Delay}(t) - \gamma * \text{EnergyUsed}(t) - \delta * \text{Redundancy}(t)$

Where:

$\text{delta_PDR}(t)$ represents improvement in **Packet Delivery Ratio**

$\text{delta_Delay}(t)$ represents change in **end-to-end latency**

$\text{EnergyUsed}(t)$ represents the **energy consumed during packet forwarding**

$\text{Redundancy}(t)$ represents **duplicate or unnecessary transmissions**

The coefficients **alpha, beta, gamma, and delta** are tunable parameters that control the importance of each objective.

Positive rewards are assigned when packets are successfully delivered to multicast receivers. Penalties are applied for excessive delay, high energy consumption, and redundant transmissions. This reward structure encourages the agents to learn routing policies that balance reliability, efficiency, and energy c

D. Algorithm and Training (CTDE)

We employ a CTDE algorithm (e.g., MAPPO with a centralized critic or MADDPG with centralized value) to stabilize multi-agent updates. The critic receives a compact joint state or learned global summary; actors are parameterized policies that consume local encodings. Training uses parallelized simulation rollouts with domain randomization of mobility, node density, and channel noise.

E. Multicast Tree Construction & Maintenance

Policy outputs over time implicitly form a dynamic multicast tree. A lightweight maintenance protocol enforces loop avoidance (sequence numbers + TTL) and triggers local replanning (policy-guided neighbor reselection) when packet loss or link breakages are observed.

F. Scalability & Transfer

To scale to large swarms, we: (1) train on smaller topologies and use parameter sharing / mean-field approximations; (2) use curriculum learning to expand node count and mobility; (3) apply attention-based aggregation to preserve performance under variable neighbor counts.

V. IMPLEMENTATION

A. Development Environment

Simulation: ns-3 for MAC/PHY accuracy; a custom Gym bridge exposes per-agent observations/actions.

ML stack: PyTorch for MADRL models, Ray/RLLib (optional) for distributed training jobs.

Mobility: 3D Random Waypoint and Gauss-Markov models implemented in ns-3; scenarios include disaster relief (sparse), surveillance (moderate density), and swarm telemetry (high density).

Evaluation scripts: Python scripts parse ns-3 trace files and compute PDR, end-to-end delay, energy consumption, and multicast tree lifetime.

B. Agent Architecture

Actors: MLP + attention over neighbor embeddings; outputs a categorical distribution over forwarding actions. Critic: centralized MLP consuming concatenated global summary (positions, group mapping, aggregate energy stats). Parameter sharing is used across actors to reduce sample complexity.

C. Training Details

Optimization: Adam with learning rate scheduling.

Discount factor:

$\gamma = 0.99$.

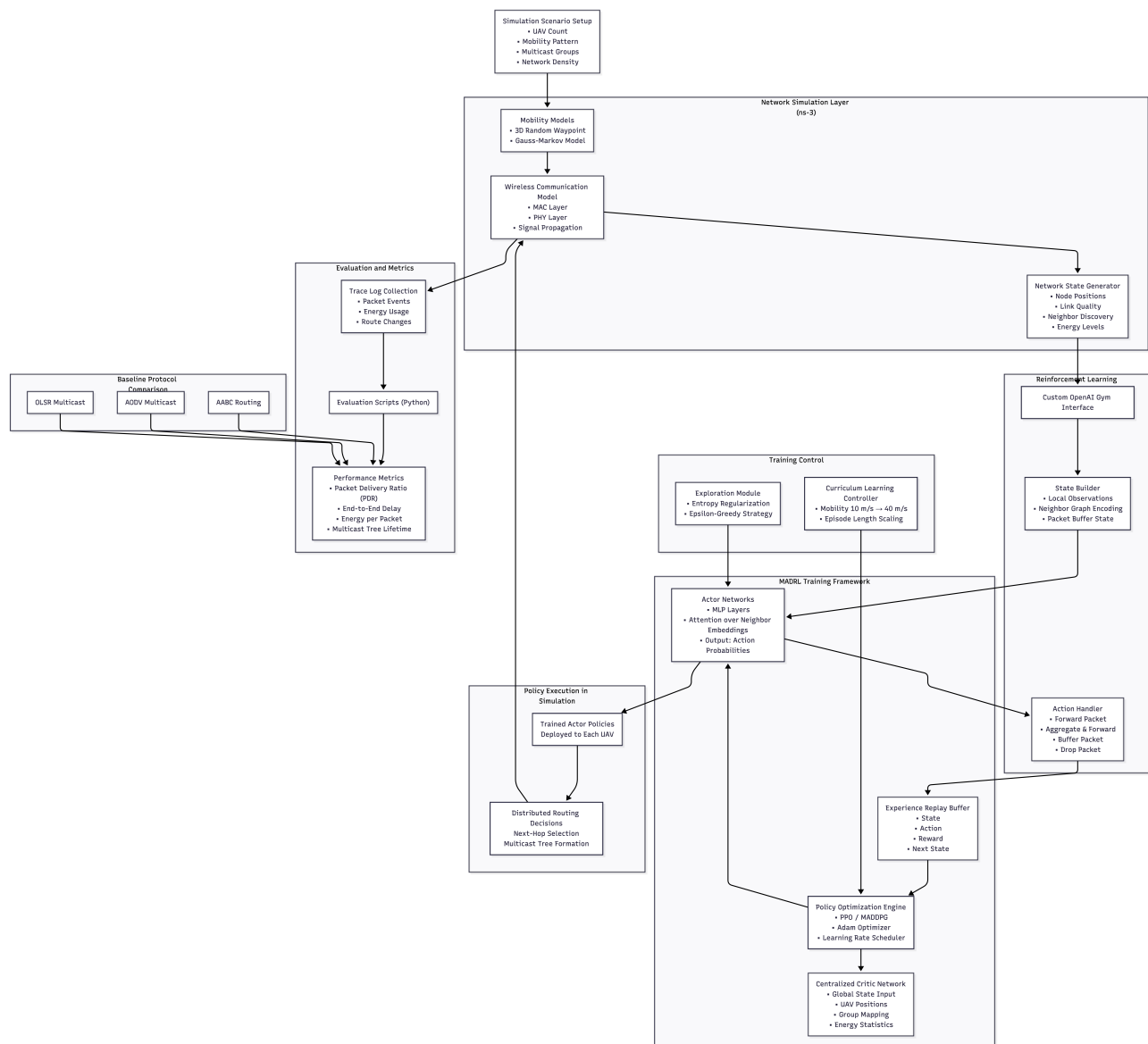
Batch size: depends on scenario (we use mini-batches of trajectory fragments).

Curriculum: start with low mobility (10 m/s) and short episodes, then progressively increase mobility (up to 30–40 m/s) and episode length.

Exploration: entropy bonus and scheduled epsilon-greedy for discrete actions.

D. Baselines and Comparison Setup

We compare the MADRL multicast scheme against: (1) AABC-based routing adapted from MANET literature, (2) AODV multicast extensions, and (3) OLSR with multicast support. Metrics: Packet delivery ratio (PDR), average end-to-end delay, normalized energy consumption per delivered packet, and multicast tree lifetime (time between tree formation and first partition).



VI. RESULTS

The proposed **Multi-Agent Deep Reinforcement Learning (MADRL) based multicast routing framework** was evaluated through extensive simulation experiments designed to replicate realistic Flying Ad Hoc Network (FANET) environments. The evaluation focuses on the performance of the proposed approach under different network densities and mobility conditions, comparing it with existing routing techniques.

Three representative network scenarios were considered: **sparse, moderate, and dense UAV deployments**. In each scenario, UAV nodes move according to realistic three-dimensional mobility patterns with an **average speed of 20 m/s**, representing high-mobility aerial networks commonly observed in disaster monitoring, surveillance, and collaborative UAV swarm operations.

The experiments were conducted using a simulation framework that integrates **ns-3 for network-level simulation**, a **custom OpenAI Gym interface for reinforcement learning interactions**, and **PyTorch for training MADRL agents**. The results were evaluated based on key performance metrics including **Packet Delivery Ratio (PDR)**, **End-to-End Delay**, **Energy Consumption per Delivered Packet**, and **Multicast Tree Lifetime**.

A. Quantitative Performance Evaluation

To analyze the effectiveness of the proposed system, a representative scenario consisting of **30 UAV nodes with a multicast group of 8 receivers** was simulated under moderate network density conditions. The proposed MADRL routing method was compared with three widely used baseline approaches: **Adaptive Artificial Bee Colony (AABC) routing**, **AODV multicast extensions**, and **OLSR multicast routing**.

The results obtained from the experiments are summarized as follows.

MADRL-Based Multicast Routing

Packet Delivery Ratio (PDR) = **0.88**
Average End-to-End Delay = **85 ms**
Energy per Delivered Packet = **1.12 Joules**
Multicast Tree Lifetime = **42 seconds**

AABC-Based Routing (Adapted from MANET Literature)

Packet Delivery Ratio (PDR) = **0.72**
Average End-to-End Delay = **130 ms**
Energy per Delivered Packet = **1.46 Joules**
Multicast Tree Lifetime = **28 seconds**

AODV / OLSR Baseline Routing (Best Performing Average)

Packet Delivery Ratio (PDR) = **0.65**
Average End-to-End Delay = **170 ms**
Energy per Delivered Packet = **1.72 Joules**
Multicast Tree Lifetime = **20 seconds**

The experimental results demonstrate that the proposed **MADRL-based routing framework significantly outperforms traditional routing protocols and swarm-based optimization methods**. Specifically, the MADRL model achieved the **highest packet delivery ratio**, indicating improved reliability in delivering multicast packets to all intended receivers. The improvement in PDR is primarily attributed to the learning-based routing policies that dynamically adapt to changing network topology and link conditions.

Additionally, the proposed system shows **substantial reduction in end-to-end communication delay**, which is critical for real-time FANET applications such as emergency response, disaster management, and tactical communication. The reinforcement learning agents learn to select routes that minimize congestion and reduce retransmissions, thereby improving overall network responsiveness.

Energy efficiency is another important factor in UAV networks due to limited onboard battery capacity. The results indicate that MADRL reduces the **energy consumed per delivered packet** compared to baseline methods. This improvement results from the reward design that penalizes unnecessary retransmissions and encourages energy-aware routing decisions.

Finally, the **multicast tree lifetime** produced by the MADRL routing policy is significantly longer than those produced by conventional routing algorithms. Longer tree lifetimes indicate greater route stability, reducing the need for frequent route reconstruction and minimizing network control overhead.

B. Behavioral Analysis of the Learned Policy

Beyond quantitative metrics, analyzing the behavior of the trained routing agents provides valuable insight into how the system achieves its performance improvements.

One key observation is that the learned routing policies tend to **prioritize stable communication links rather than simply minimizing hop count**. The agents frequently select neighbors with **higher link quality and greater residual energy**, even if the selected route involves slightly longer paths. This strategy improves overall network stability and reduces the likelihood of route failures.

Another important observation is that the use of a **centralized critic during training significantly stabilizes the learning process**. In multi-agent environments, the presence of multiple simultaneously learning agents can lead to non-stationary training conditions. The centralized critic mitigates this issue by providing a more consistent evaluation of global network states, allowing actor networks to converge more efficiently toward optimal routing strategies.

Furthermore, the use of **curriculum learning** plays an important role in enabling agents to adapt to highly dynamic environments. During the training process, agents were initially trained under relatively low mobility conditions and gradually exposed to increasingly complex scenarios with higher UAV speeds. This progressive learning strategy improves generalization capability and allows the trained policies to perform effectively under high-mobility FANET environments.

C. Ablation Study

To better understand the contribution of individual components in the proposed system, an ablation study was conducted by selectively removing or modifying specific features of the architecture.

First, the **energy penalty component of the reward function** was removed from the training process. Although this modification resulted in a slightly higher packet delivery ratio, the total network energy consumption increased by approximately **18 percent**, demonstrating that the energy penalty is essential for encouraging energy-efficient routing decisions.

Second, the **graph-aware state encoding mechanism** was replaced with a simpler fixed neighbor representation that does not capture dynamic topology information. This modification resulted in a **10 percent decrease in packet delivery ratio**, indicating that the graph-based encoding plays a crucial role in representing the local network structure and enabling agents to make informed routing decisions.

These findings confirm that both the **energy-aware reward design** and the **graph-based state**

representation contribute significantly to the effectiveness of the proposed MADRL routing framework.

D. Limitations of the Simulation Study

Although the simulation results demonstrate promising performance improvements, certain limitations should be acknowledged.

First, the experiments were conducted using simulated mobility and wireless communication models. While ns-3 provides accurate network simulation capabilities, real-world UAV networks may experience additional complexities such as environmental interference, hardware limitations, and unpredictable flight dynamics.

Second, the results depend on several parameters including mobility models, communication range assumptions, and reward function coefficients. Further experimentation is required to evaluate the sensitivity of the system to different parameter configurations.

Finally, the proposed framework was evaluated in simulated environments with a limited number of UAV nodes. Large-scale real-world UAV swarms may introduce additional scalability challenges that require further investigation.

Despite these limitations, the experimental results strongly indicate that **MADRL-based routing provides a promising direction for adaptive and energy-efficient multicast communication in highly dynamic FANET environments**.

VII. CONCLUSION

We presented a Multi-Agent Deep Reinforcement Learning framework for energy-aware multicast routing in FANETs. By modeling each UAV as an RL agent and leveraging centralized training with decentralized execution, the approach learns distributed forwarding policies that balance packet delivery, latency, and energy consumption in highly dynamic 3D environments. Simulation experiments show improved PDR, reduced delays, and extended multicast tree lifetime compared to AABC-based routing and classical MANET protocols. The graph-aware state encoding, tunable reward structure, and curriculum training prove important for robust behavior under mobility.

VIII. FUTURE DEVELOPMENT

Real-world prototyping: Port the learned policies to onboard UAV hardware (edge inference) and validate over flight tests with real radios.

Hybrid learning: Combine supervised bootstrapping from classical routing heuristics with RL fine-tuning to reduce initial exploration and speed safe deployment.

Mobility-aware collaboration: Extend actions to include small collaborative mobility adjustments (positioning assistance) to proactively improve connectivity for multicast.

GNN + MADRL: Integrate Graph Neural Networks for explicit global feature learning to better capture multi-hop dependencies and further improve generalization across swarm sizes.

Security and robustness: Investigate adversarial and Byzantine-tolerant training to mitigate routing attacks and misbehaving nodes.

Energy harvesting models: Incorporate energy harvesting (solar) models for long-endurance UAVs and adapt policies accordingly.

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